

# 5940 2024 “Roti” Technical Binder



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# Game Analysis

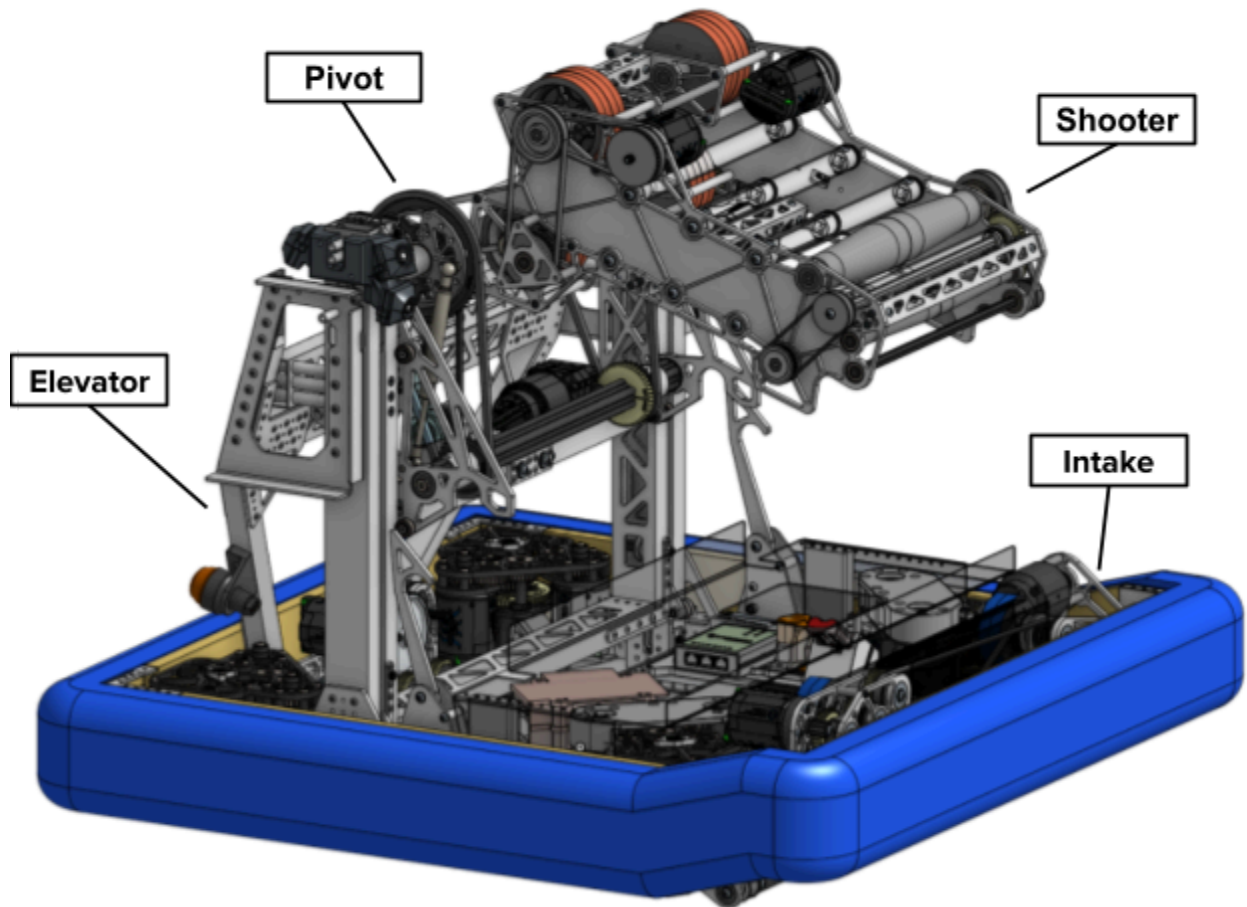
The 2024 FRC game, Crescendo, presents several challenges: scoring Notes in the Amp, Speaker, Trap, and lifting the robot onto a chain to get Onstage. During kickoff weekend, we created a list of requirements for our robot:

## Qualifications

Ranking point criteria:

- 2 points: Win the match
  - 1 point: Score 18 Notes during the teleoperated period into the speaker and the amp. This threshold decreases to 15 if both alliances use the coopertition bonus.
  - 1 point: Score 10 points in the endgame and have 2 robots Onstage (includes Onstage, Spotlight, and Trap points)
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- Score in the Amp
  - Be able to score in the Trap to increase the likelihood of getting the Endgame bonus Ranking Point
  - Be short enough to drive under the Stage to avoid defenders and generally avoid traffic
  - Shoot over maximum legal height (48") defenders from the Podium into the Speaker

# Robot Design

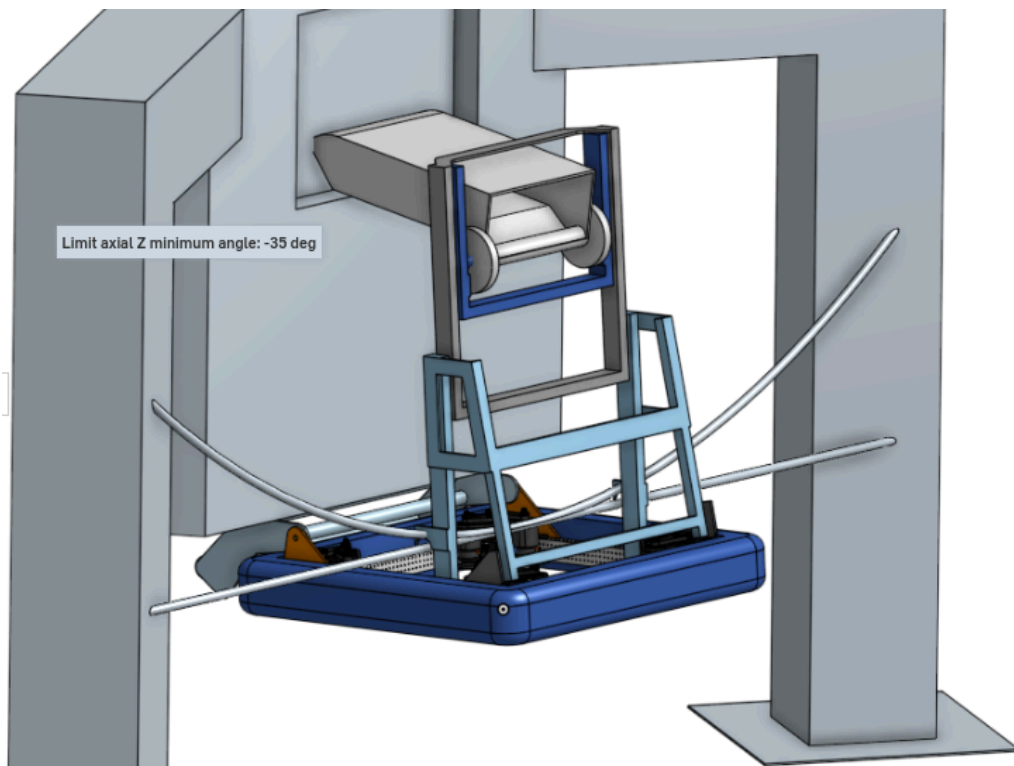


## Robot Architecture

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Roti is a “shooter-on-elevator” architecture. The main reasons why we went for this concept are:

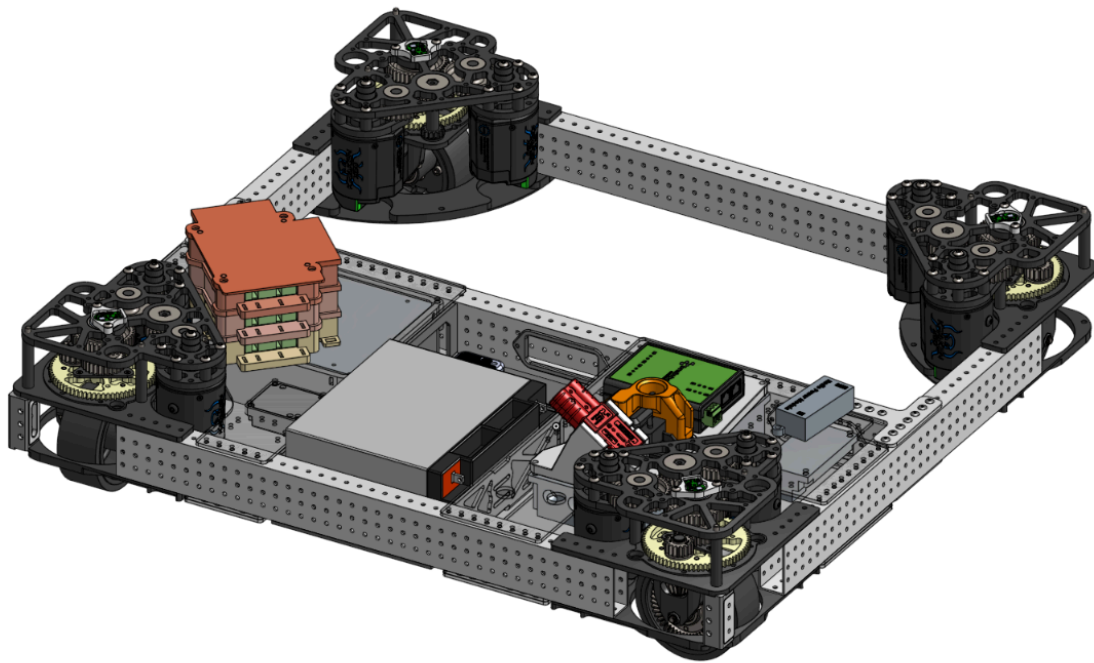
- The elevator allowed us to use one mechanism (the shooter) to score in all possible locations (Speaker, Amp, Trap). Additionally, the elevator can be used to pull the robot onto the chain.
- Since the shooter is on top of the elevator, we can drive under the Stage when the elevator is down, and shoot over defenders by extending the elevator to full height



Early “Crayon CAD” of our robot concept

## Drivebase

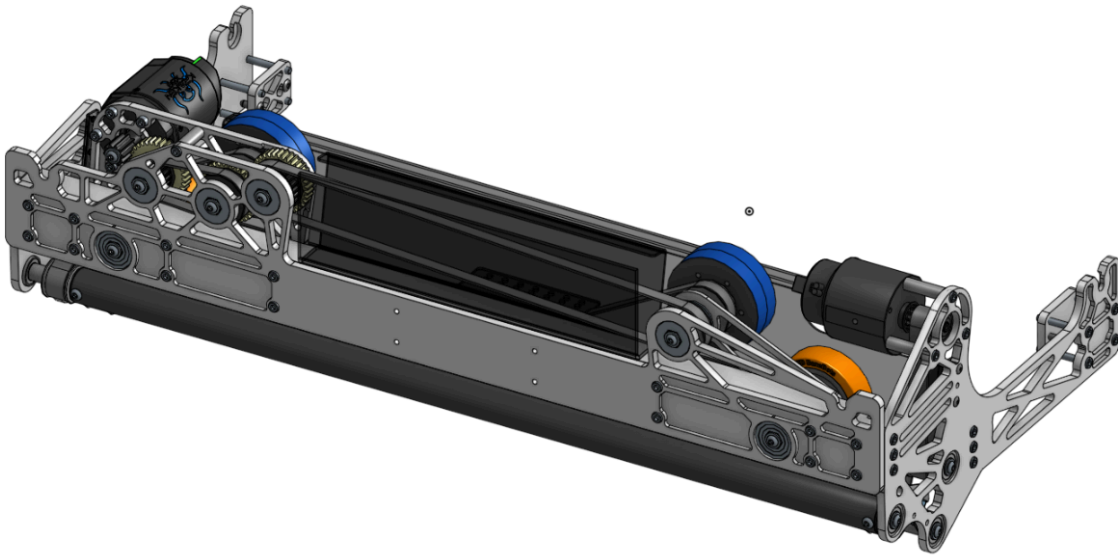
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- Powered by WCP Swerve X inverted swerve modules with Kraken X60 motors
- X3 gear ratio set with 10T motor pinion produces a 19 ft/s free speed
- Wheelbase is the same as a standard swerve robot with a 26" x 28" frame perimeter; with the intake, the robot is 30.5" x 28"

## Intake

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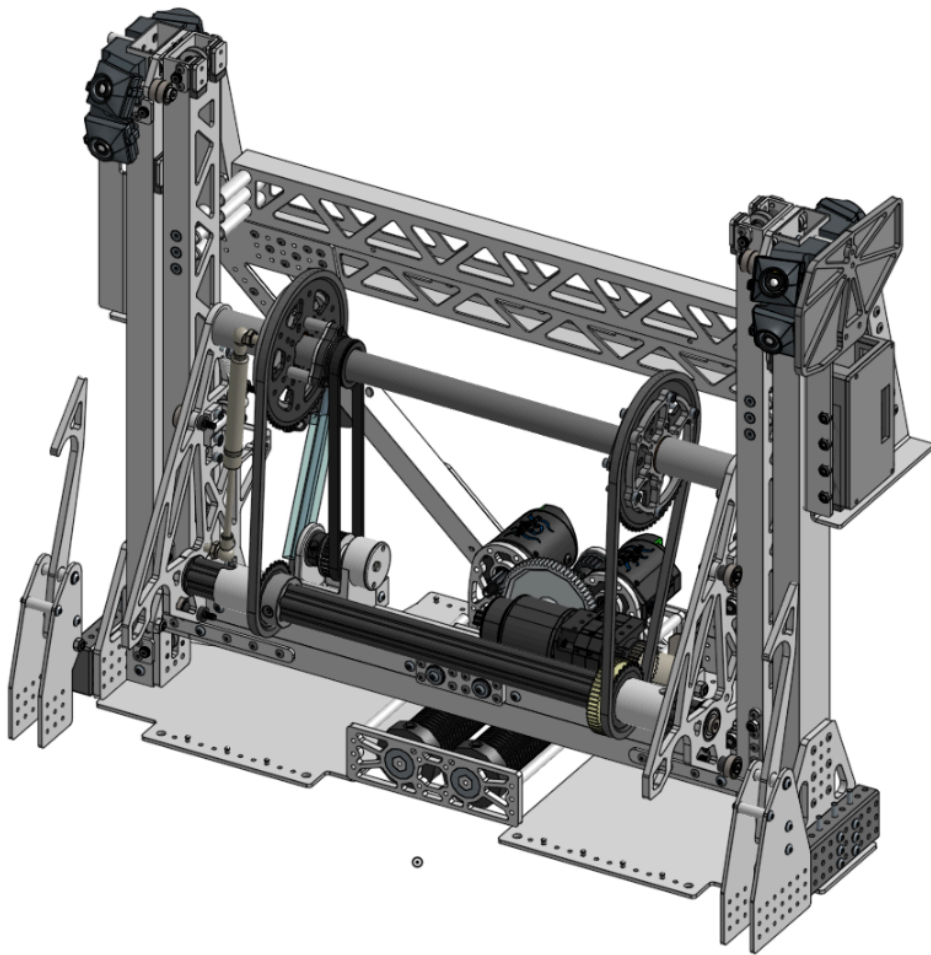
The Ground Intake collects Notes from the ground and centers them before they enter the Shooter.

- Under-the-bumper intake
  - Full width (28" wide)
  - Makes drivetrain only 4.5" longer
- 3x 1" OD aluminum rollers covered with silicone tubing grab Notes off the ground
  - 1.3 : 1 reduction
  - 19.6 ft/s surface speed
- 4x 3+4" wheels center Notes
  - 4.8 : 1 reduction
  - 16.4 ft/s surface speed



# Elevator

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The elevator translates the shooter up and down, and is also used to get the robot Onstage.

## Extension

- 2 stage continuous elevator with overextending carriage
- Internally rigged using UHMWPE rope
- Crossbars on the base and first stage are lowered so that the robot can shoot into the speaker while staying below the height of the stage

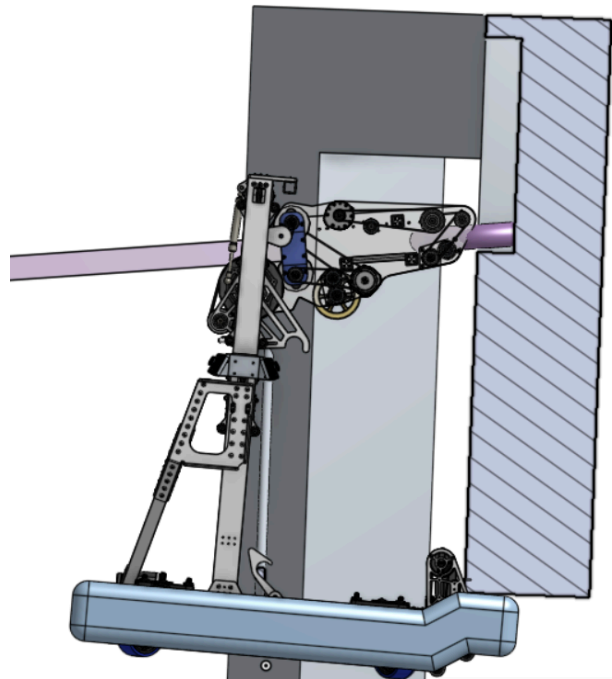
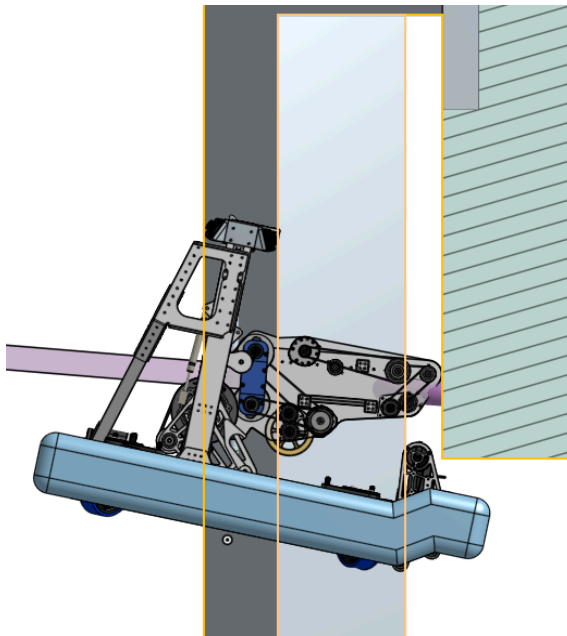
## Gearbox

- Driven by 2x Kraken X60s at a 10.37 : 1 reduction
- 1.5" diameter 3d printed spools



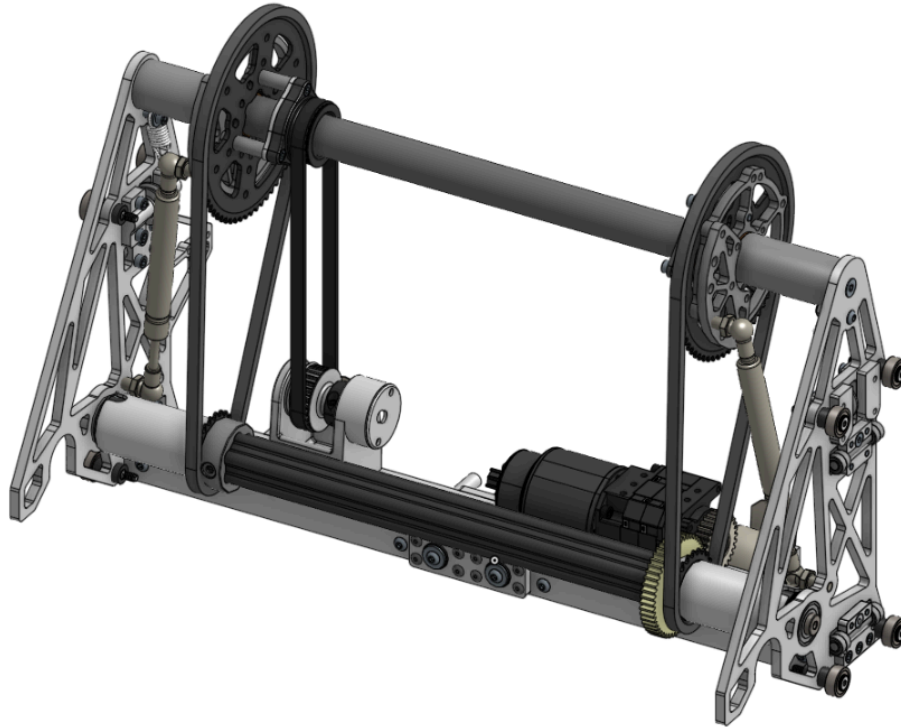
## Climbing Sequence

- Hooks on carriage pull the robot off the ground
- Once the chain is pulled far enough into the robot, an additional set of hooks located near the bottom of the robot passively latch on
- Once the passive hooks engage, the elevator is free to extend to full height in order to score a Note in the Trap.
- Robot center-of-gravity is located so the robot constantly wants to tip toward the trap



## Pivot

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The shooter pivot is part of the elevator 2nd stage / carriage, and adjusts the angle of the shooter in order to score in the Speaker, Amp, and Trap.

### Structure

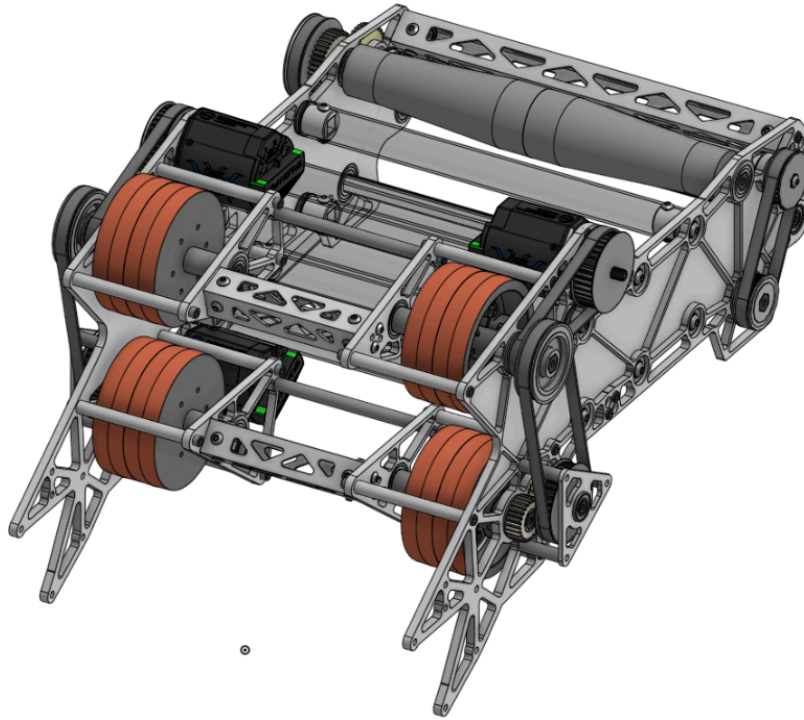
- 1" OD 1/16" wall aluminum tube is used as a dead axle
- Chain climbing hooks are integrated into the pivot side plates

### Gearbox

- 114.58 : 1 overall gear reduction
  - 25 : 1 Maxplanetary gearbox provides a high gear reduction in a compact space
  - Final reduction stage uses 2x #25 chain runs
- 2 x 50 lb gas springs counteract the torque created by the shooter's weight about the pivot and aid controllability by providing dampening
- CANcoder belted directly to the shooter pivot shaft gives accurate measurements of the shooter's angle

# Shooter

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The Shooter scores Notes in the Speaker, Amp, and Trap

## Shooter Wheels

- 2 Kraken X60s spin 2 sets of shooter wheels each
  - 4" Shooter wheels driven with a 2 : 1 upduction
  - Wheels on opposite sides of the shooter are spun at different speeds to impart spin on the Note as it leaves the shooter

## Feeder System

- 3d printed conical wheels and mecanum wheels help spread out note inside the shooter, improving shot consistency
- Front roller pair angled downward relative to the shooter exit path to score more effectively in the Trap
- 6x 1" OD polycarbonate idler rollers reduce friction on the Note while it moves through the shooter